



JavaScript Unit Testing

Writing unit tests is as important as writing the code itself: it helps to ensure that the code is written according to a given specification and that it remains correct as it evolves.

Testing Framework

Testing the code starts with a testing framework. The framework provides a level of abstraction that makes it possible to write tests in an easy and efficient way. It also provides a set of tools to run the tests, make assertions and report the results.

Odoo developers use a home-grown testing framework called `HOOOT`. The main reason for using a custom framework is that it allows us to extend it based on our needs (tags system, mocking of global objects, etc.).

On top of that framework we have built a set of tools to help us write tests for the web client (`web_test_helpers`), and a mock server to simulate the server side (`mock_server`).

You can find links to the reference of each of these parts below, as well as a section filled with examples and best practices for writing tests.

Setup

Before learning how to write tests, it is good to start with the basics. The following steps will ensure that your test files are properly picked up by the test runner.

Note that in existing addons, most of these steps can be skipped since the proper folder structure and asset bundles are probably set up.

1. Writing files in the right **place**:

All JavaScript test files should be put under the `/static/tests` folder of the related addon (e.g. `/web/static/tests/env.test.js`).

2. Using the right **name**:

Test files must end with `.test.js` . This is not only a convention, but a requirement for test files to be picked up by the runner. All other JavaScript files will be interpreted either as

Note

It is to be noted that there is an exception for `.hoot.js` files, which are not considered as test files, but as global modules for the whole test run, while other JavaScript modules are re-created for each test suite. Since the same instance of these modules will be running for the whole test run, they follow strict constraints, such as restricted imports, or advanced memory management techniques to ensure no side-effects are affecting tests.

3. Calling the files in the right **bundle**:

Test files, added in the right folder, must be included in the `web.assets_unit_tests` bundle. For ease of use, this can be done with glob syntax to import all test and test helper files:

```
# Unit test files
'web.assets_unit_tests': [
    'my_addon/static/tests/**/*.js',
],
```

4. Heading to the right **URL**:

To run tests, you can then go to the `/web/tests` URL.

Tip

This page can be accessed through  **Debug menu** ▶ **Run Unit Tests**.

Writing tests

After creating and including test files, it is time to write tests. You may refer to the following documentation sections to learn about the testing framework.

HOOT

Web test helpers

Mock server



Get Help

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